

Amrutam Pharma App

Ayurvedic Super-App Feature Specification & Clinical Architecture

Executive Overview

The **Amrutam Pharma App** is an enterprise-grade mobile application combining **1-on-1 Ayurvedic Telemedicine Consultations**, an **Authenticated Classical Botanical Pharmacy (20,000 items)**, and an **Encrypted AYUSH Clinical Health Record Vault (10,000 records)**. Built on a modular, offline-first React Native architecture, it implements strict clinical safety checks such as real-time **Allergy Collision Detection** and **Optimistic Slot Concurrency Protection** against double-booking.

Module / Layer	Core Responsibilities	Scale / Datasets
Consultations	Physician discovery, slot reservation with concurrency tokens, CCIM credential verification, and 1-on-1 encrypted video teleconsultation.	5,000 Doctors 8 slots/doctor daily
Ayurvedic Shop	2-column virtualized infinite scroll, classical treatise references (Charaka Samhita), Dravyaguna pharmacology, allergy collision warnings, and offline cart.	20,000 Products 7 categories
Health Records	Chronological medical timeline grouped by month/year with SVG connectors, pulse diagnostics (Nadi Gati), PDF report downloads, and one-tap prescription reorder.	10,000 Records Prescriptions & Labs
Dosha Quiz & Profile	5-question clinical Prakriti assessment (Vata / Pitta / Kapha) scoring Tridosha balance and personalizing doctor and remedy recommendations across tabs.	Tridosha Engine Cross-module state
Offline Sync Engine	Zero-backend in-memory server, NetInfo live connectivity monitoring, Redux persist whitelist, and offline action synchronization queue.	Offline-First Auto-reconciliation

1. Module 1 — Clinical Consultations & Telemedicine

- **Optimistic Slot Concurrency Protection:** Slots are modeled with unique version tokens. When a patient books a slot, the client evaluates local status while the server validates the version token, preventing double-booking and stale reservations under concurrent load.
- **Central Council of Indian Medicine (CCIM) Registrations:** Doctor cards feature verified practitioner badges, registration numbers (e.g. CCIM/UP/AYUR/8492), institutional degrees (BHU, Gujarat Ayurved University), and OPD schedules.
- **1-on-1 HD Video Room:** Encrypted video consultation room with live pulse assessment status, camera/mic controls, PiP self-view, and automated clinical prescription creation.

2. Module 2 — Ayurvedic Pharmacy & Cross-Module Allergy Safety

- **20,000 Item Infinite Virtualization:** Optimized for instant scrolling with unique high-resolution botanical CDN image assets.
- **Dravyaguna Pharmacology:** Complete Ayurvedic scientific breakdown including Rasa (Taste), Guna (Qualities), Virya (Potency: Sheeta/Ushna), and Vipaka (Post-digestive metabolic effect).
- **Cross-Module Allergy Warning Engine:** The Shop module actively reads logged allergy histories from state.healthRecords.records (e.g., Peanuts, Gluten, Dairy, Synthetic Preservatives). If a remedy contains conflicting

ingredients, prominent danger badges appear on cards and product detail views.

- **Anupana Carrier Directions:** Detailed clinical vehicle recommendations (e.g. warm milk at bedtime, lukewarm water with honey).

3. Module 3 — AYUSH Health Records & Prescription Reordering

- **Chronological Medical Timeline:** 10,000 records structured chronologically with custom SVG TimelineConnector node trees and month/year section headers.
- **Nadi Pariksha Diagnostics:** Bio-rhythm pulse analysis classifying wave velocities into classical *Sarpagati* (Pitta - serpentine), *Jalaukagati* (Vata - leech-like), and *Hamsagati* (Kapha - swan-like).
- **One-Tap Prescription Reordering:** Direct cross-module bridge allowing patients to click '**Reorder Prescribed Medicines**' on any prescription to automatically prefill the Shop cart with accurate items and dosage regimens.

4. Ayurvedic Medical Design System & SVG Visual Language

The app employs an Ayurvedic aesthetic system adhering to strict medical UX principles:

Design Primitive	Component File	UX Purpose & Clinical Aesthetic
Z-Axis Hero Banner	ZAxisHeroBanner.tsx	2-layer opacity crossfade with scale depth illusion and SVG linear gradient scrim.
Wave Divider	WaveDivider.tsx	Soft curved organic section breaks between doctor profiles and headers.
Dashed Divider	DashedDivider.tsx	Cross-platform crisp SVG dashed rules for bill summaries and filter sheets.
Progress Ring	ProgressRing.tsx	Circular SVG capacity meters displaying today's doctor slot availability.
Timeline Connector	TimelineConnector.tsx	Vertical dotted line and color-coded node chaining health records chronologically.
Badge Seal	BadgeSeal.tsx	Circular dashed stamp certifying 100% Ayurvedic authenticity & CCIM physician status.
Botanical Pattern	LeafPatternBackground.tsx	Deterministic botanical leaf backdrop for onboarding, auth, and empty states.

5. Offline-First Synchronization Architecture

- **In-Memory Zero-Backend Engine (mockServer.ts):** Full in-memory request interceptor delivering instant responses, pagination, multi-criteria filtering, and slot concurrency validation without requiring active backend instances.
- **Redux Persist Integration:** Whitelists essential client state (auth, cart, wishlist, syncQueue, healthRecords, patientProfile) into AsyncStorage.
- **NetInfo Connectivity Monitoring & Sync Queue:** Detects network drops, displays a top warning banner, queues booking and cart transactions, and reconciles pending actions upon reconnection.

6. Quality Assurance & Verification

- **TypeScript Typecheck:** npx tsc --noEmit passed with **0 errors** across the entire repository.
- **Code Preservation:** Previous chat application source safely preserved in frontend-new/src_temp/ for multimedia player reuse.